

Register are used to store and shift binary data. (consists of a group of flip-flops with common clock)

Parallel register / shift register
Counter change states in a prescribed sequence. (consists of two or more FF) $nFF \rightarrow a \text{ MOD } 2^n \text{ counter}$

Register Binary counter / Ripple counter / synchronous counter
fed from output of the previous FF $\rightarrow$ same clk to all FF

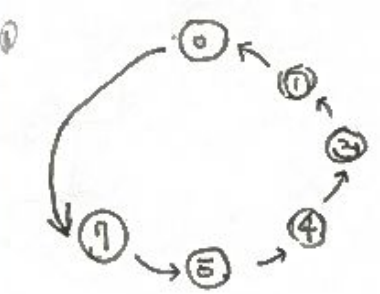
+ clk
+ Load
+ C/rN = 0 $\rightarrow$ All output = 0

$\uparrow$ 使 present value 清楚 0

Shift Register is a register in which binary data can be stored, and this data can be shifted to the left or right when a shift signal is applied.

shift = 1 $\rightarrow$ clk enabled $\rightarrow$ shift
shift = 0 $\rightarrow$ no shift occurs

Synchronous Binary counter (0 $\rightarrow$ 7)



step 1: 決定幾個 FF

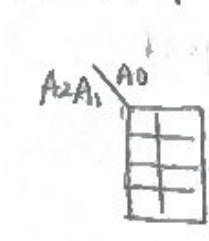
($n \text{ FF} \rightarrow 2^n \text{ counter}$)
6 counter, 3 FF

step 2: State diagram & circuit

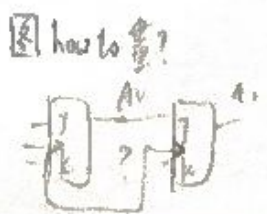
excitation table

Present	Next State	FF inputs
$A_2 A_1 A_0$	$A_2^+ A_1^+ A_0^+$	T_A, T_A, T_A
0 0 0	0 0 1	0 0 0
0 0 1	0 1 0	0 0 0
0 1 0	0 1 1	0 0 0
0 1 1	1 0 0	0 0 0
1 0 0	1 0 1	0 0 0
1 0 1	1 1 0	0 0 0
1 1 0	1 1 1	0 0 0
1 1 1	0 0 0	0 0 0

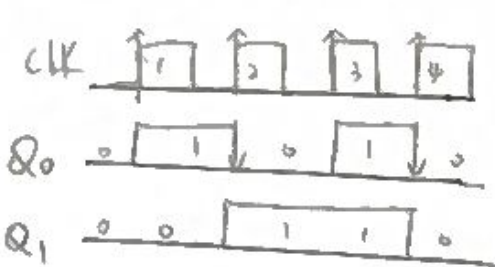
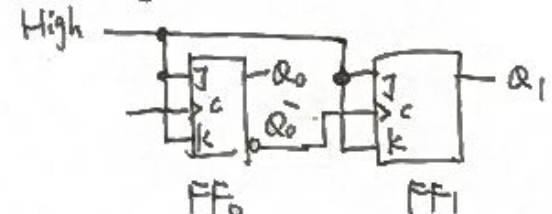
step 3: K' map & draw diagram



$T_{A2} = \sim$
 $T_{A1} = \sim$



Asynchronous (Ripple) Counters



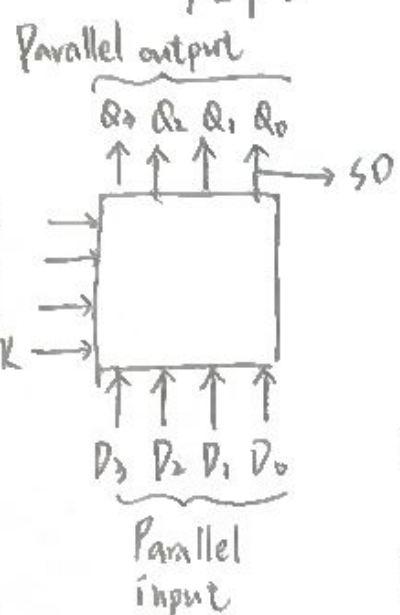
ch 13 4.1.1
(2) (5) (6)

(3) Register
a window

(17) sliding window

$E_n = 1$ & Load = 1 $\rightarrow$ 會收到值

SISO
SIPO: serial input parallel output
PISO
PIPO



Inputs		Next State				Action
Sh	L	Q_3^+	Q_2^+	Q_1^+	Q_0^+	
0	0	Q_3	Q_2	Q_1	Q_0	No change
0	1	D_3	D_2	D_1	D_0	Load
1	X	SI	Q_3	Q_2	Q_1	Right-shift

ch 14
NRZ: 完全不變
NRZI: 遇到 $X=1$, 狀態變成(前一狀態)
 $X=0$, no change

RZ: 遇到 $X=1$, 前半 bit 秒 = 1
後半 bit 秒 = 0
 $X=0$, 整秒 0

Manchester code: $X=1$, 前半 bit 秒 = 1
後半 bit 秒 = 0
 $X=0$, 前半 bit 秒 = 0
後半 bit 秒 = 1

